

## A Climate Resilience Pathway Between India and China



**For Prelims:** El Niño , Hindu Kush Himalaya (HKH) , Glacial Lake Outburst Floods (GLOFs) , Brahmaputra , India-China Strategic Economic Dialogue , Green Grids Initiative - One Sun One World One Grid , Green and Sustainable Development Partnership , International Solar Alliance , India-EU Clean Energy and Climate Partnership

**For Mains:** Climate Change as a New Dimension of India-China Relations, Climate Diplomacy and India's Foreign Policy

Source: TH

### Why in News?

Experts have highlighted that shared climate vulnerabilities could serve as a pragmatic, low-risk avenue for **India-China climate cooperation** , following **El Niño-delayed monsoons** and **extreme flooding** in Indian states such as **Assam and Odisha** and Chinese provinces including **Shaanxi, and Gansu** .

- Further, the **April 2026 visit of China's Special Envoy for Climate Change to New Delhi** has reinforced climate resilience and disaster management as potential areas of bilateral engagement despite broader geopolitical tensions.

### Summary

- **India and China face common climate challenges** , including **extreme weather events, glacial retreat, floods, and heatwaves** , making **climate resilience and disaster management** potential areas for constructive cooperation despite geopolitical tensions.
- **Strengthening cooperation through scientific collaboration, data sharing, urban resilience, climate finance, and multilateral platforms such as BRICS** can help both countries enhance climate adaptation while rebuilding mutual trust.

# What are the Shared Climate Vulnerabilities of India and China?

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- **The Urban "Concrete Trap" & Flooding:** Rapid, unchecked urbanization has paved over **wetlands and natural floodplains** in both countries.
  - Consequently, intense bursts of rainfall overwhelm outdated drainage systems, leading to severe urban waterlogging as witnessed in both **Mumbai and China's southern Guangxi region**.
- **The Third Pole (Himalayan Ecosystem) Risks:** The shared **Hindu Kush Himalaya (HKH) region** is witnessing accelerating glacial retreat.
  - This elevates the risk of catastrophic **Glacial Lake Outburst Floods (GLOFs)** and threatens the hydrological stability of major transboundary river **systems like the Yarlung Tsangpo (Brahmaputra) and the Sutlej**.
- **Inland Heatwaves vs. Coastal Cyclones:** Inland cities in both nations are facing a vicious cycle of prolonged heatwaves and **flash floods**.
  - Meanwhile, their highly populated coastal megacities face compounding risks from storm surges, extreme precipitation, and rising sea levels.
- **Public Health and Economic Cascades:** Beyond infrastructure damage, these events disrupt critical supply chains, depress agricultural yields, and trigger severe public health crises, effectively jeopardizing the developmental gains of both economies.

## What is the Historical Precedent for India-China Climate Engagement?

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- **Institutional Cooperation (1990s-2020):** India and China have adopted several **Joint Statements, Joint Declarations, and Memoranda of Understanding (MoUs)** on **climate change and disaster management**. Notably, their **2015 Joint Climate Statement** ahead of the **Paris Climate Conference** supported an **equitable global climate agreement** and expanded cooperation in **renewable energy, electric mobility, low-carbon urbanisation, climate adaptation, joint R&D, data sharing (hydrological, oceanic, and seismic), and disaster governance**.
  - As founding members of the **BASIC Group (Brazil, South Africa, India and China)**, both countries have coordinated their positions in global climate negotiations, advocating climate justice,
- **Strategic Economic Dialogue (SED):** The **India-China Strategic Economic Dialogue**, facilitated collaboration on **sustainable urban planning, waste management, sewage treatment, water-use efficiency, and capacity building**.
- **Sister City Agreements:** Practical cooperation was envisaged through **the Delhi-Beijing (2013), Mumbai-Shanghai (2014), and Chennai-Chongqing (2015) sister city agreements** to promote collaboration in **urban governance, environmental management, and climate resilience**.

## What are the Challenges Constraining India-China Climate Cooperation?

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- **Strategic Mistrust:** Border tensions since the **2020 clashes** have eroded trust, disrupting climate cooperation and **transboundary hydrological data sharing**, particularly on the **Brahmaputra River**.
- **Economic Competition:** China's dominance in solar manufacturing, batteries, and critical

**minerals** , coupled with India's **Atmanirbhar Bharat** strategy and a **record bilateral trade deficit of USD 112.6 billion (2025-26)** , has made India cautious about increasing dependence on Chinese green technologies, limiting bilateral cooperation.

- However, India has gradually softened its stance, as reflected in the recent easing of restrictions on Chinese firms to facilitate selective economic and investment cooperation.
- **Regulatory Barriers:** India's **2020 Press Note 3** restrictions on Chinese investments and China's export controls on **critical minerals and processing technologies** have constrained green technology collaboration and investment.
- **Weak Implementation:** Despite **Sister City Agreements**, most initiatives saw limited implementation due to diplomatic tensions and the absence of strong institutional partnerships.
- **Carbon Market Cooperation:** In **2026** , **China, Brazil, and the European Union** became the founding members of the **Open Coalition for Carbon Markets** . **India was not a founding member** , although the coalition is an **open and voluntary platform** .
- **Divergent Climate Leadership:** India has not joined China's **BRI International Green Development Coalition** , while China is not a member of India's flagship climate platforms such as the **International Solar Alliance (ISA)** and the **Global Biofuels Alliance (GBA)**, reflecting their divergent approaches to global climate leadership.

# What are India's Major Bilateral Climate Partnerships Across the World?

- **US:** Under the **India-US Climate and Clean Energy Agenda 2030** , both countries cooperate on **clean energy, climate finance, hydrogen, electric mobility, and clean technology innovation** .
- **United Kingdom (UK):** India and the UK collaborate through the **Green Growth Equity Fund (GGEF) (India's first climate-focused joint venture fund)** to mobilise private investment in **renewable energy, energy efficiency, and electric mobility** , and jointly lead the **Green Grids Initiative - One Sun One World One Grid (GGI-OSOWOG)** for global solar energy connectivity.
- **Japan:** Through the **Japan-India Clean Energy Partnership (CEP)** and the **Joint Crediting Mechanism (JCM)** , both countries collaborate on **electric vehicles (EVs), hydrogen and ammonia technologies, battery storage, industrial energy efficiency, and carbon credit mechanisms** .
- **Germany:** Through the **Green and Sustainable Development Partnership (GSDP)** (Germany committed €10 billion by 2030 to support India's eco-infrastructure) and the **International Climate Initiative (IKI)** , Germany supports India in **renewable energy, climate-resilient infrastructure, biodiversity conservation, and climate adaptation** .
- **Denmark:** The **India-Denmark Green Strategic Partnership (2020)** focuses on **offshore wind, renewable energy, circular economy, water management, and sustainable urban development** .
- **France:** India and France jointly launched the **International Solar Alliance (ISA)** at **COP21 (2015)** and collaborate on **solar energy, blue economy, sustainable marine development, and global climate action** .
- **European Union (EU):** Through the **India-EU Clean Energy and Climate Partnership** , both sides cooperate on **clean energy transition, green finance, low-carbon development, carbon markets, and climate resilience** .

## Strategic Significance for India

- **Bridging the Climate Finance Gap:** India's ambitious **Nationally Determined Contributions (NDCs)** require trillions of dollars. Bilateral partnerships unlock vital concessional loans, blended finance models, and green bonds.
- **Critical Technology Transfer:** Developed countries possess highly advanced technologies in offshore wind, green hydrogen, and battery storage.
  - Partnerships with nations like Denmark and Germany facilitate the transfer and localization of these critical technologies.
- **Geopolitical Leverage:** By forming strategic green alliances with Western powers, India consolidates its position as a responsible global climate leader, effectively advocating for the Global South and defending the principle of **Common but Differentiated Responsibilities (CBDR)**.

# How can India and China Strengthen Cooperation on Climate Resilience?

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- **Focusing on "Low-Politics" Entry Points:** India and China must explicitly separate disaster management from strategic border disputes.
  - Bilateral dialogues should begin strictly with non-competitive, actionable sectors such as sustainable agriculture, urban waste management, and early warning systems before attempting to negotiate highly competitive green tech supply chains.
- **Synergizing Subnational Best Practices:** Cooperation should bypass national gridlock by reviving subnational (city-to-city) knowledge transfers.
  - India's municipalities can adopt China's data-driven "**Sponge Cities**" model (using permeable pavements and green infrastructure to absorb rainwater), while China can learn from India's highly successful, community-led **Heat Action Plans (like the Ahmedabad model)**.
- **Leveraging Minilateral Platforms:** Direct bilateral engagement can be fragile. Utilizing minilateral development platforms like **BRICS** and the **Asian Development Bank (ADB)** can provide a neutral, structured environment to negotiate technical and financial flows without triggering bilateral geopolitical sensitivities.
- **Institutionalizing Blended Climate Finance:** To bridge the massive adaptation finance gap, both governments should collaboratively develop frameworks for **blended finance** and green municipal bonds.
  - By establishing common fiscal standards, India and China can **de-risk private investments** and jointly advocate for fairer climate finance architecture for the Global South at multilateral forums.
- **Himalayan Hydrological Surveillance:** Moving beyond erratic seasonal data sharing to **establish continuous, year-round joint glaciological monitoring** of the Hindu Kush Himalaya (HKH) region.
  - This is critical for predicting GLOFs and managing volatile transboundary river basins.

## ***Drishti Mains Question:***

Examine the role of climate diplomacy in strengthening India's bilateral relations. How can climate resilience contribute to confidence-building between India and China despite geopolitical tensions?

# Frequently Asked Questions (FAQs)

## 1. Why is climate resilience considered a potential area for India-China cooperation?

Shared vulnerabilities such as **floods, heatwaves, glacial retreat, and extreme weather events** make **disaster management and climate adaptation** relatively low-risk areas for bilateral cooperation.

## 2. What is the significance of the Hindu Kush Himalaya (HKH) region for India and China?

The **HKH region** is the source of major **transboundary rivers** and is increasingly vulnerable to **glacial retreat and Glacial Lake Outburst Floods (GLOFs)**, making joint monitoring crucial.

## 3. What are the major challenges to India-China climate cooperation?

Key challenges include **border tensions, strategic mistrust, economic competition, regulatory barriers, and weak implementation of past agreements**.

## 4. What are the India-China Sister City Agreements?

The **Delhi-Beijing (2013), Mumbai-Shanghai (2014), and Chennai-Chongqing (2015)** agreements were designed to promote cooperation in **urban governance, environmental management, and climate resilience**.

## 5. How can BRICS contribute to India-China climate cooperation?

**BRICS** can provide a **neutral multilateral platform** for collaboration on **climate finance, disaster resilience, technology sharing, and sustainable development**, reducing bilateral geopolitical sensitivities.

[Watch Video on YouTube:

► <https://www.youtube.com/embed/SHnJN2A7dUQ>

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## UPSC Civil Services Examination Previous Year Question (PYQ)

### *Prelims*

**Q. “Belt and Road Initiative” is sometimes mentioned in the news in the context of the affairs of : (2016)**

- (a) African Union
- (b) Brazil
- (c) European Union
- (d) China

**Ans: (d)**

### *Mains*

**Q. The China-Pakistan Economic Corridor (CPEC) is viewed as a cardinal subset of China’s larger ‘One Belt One Road’ initiative. Give a brief description of CPEC and enumerate reasons why India has distanced itself from the same. (2018)**

# Taxation and Other Laws (Amendment) Bill, 2026



Source: TH

## Why in News?

The Lok Sabha passed the Taxation and Other Laws (Amendment) Bill, 2026, replacing the **Income-tax (Amendment) Ordinance, 2026 and amending the Income-tax Act, 2025, Finance Act, 2026, and Payment and Settlement Systems Act, 2007.**

- The Bill seeks to attract foreign investment, promote the Make in India initiative, enhance ease of doing business, and strengthen India's digital and financial ecosystem.

## What are the Key Provisions of the Taxation and Other Laws (Amendment) Bill, 2026?

- **Push for Electronics Manufacturing:** Under the **Income Tax Act 2025**, foreign companies supplying capital goods or utilizing Indian factories for contract manufacturing of electronics were **exempt from income tax for five years.**
  - The new Bill explicitly defines eligible goods (mobile phones, laptops, servers, wearables) and extends this **tax holiday** by another ten years (up to **2040-41**).
- **Support for Electronics and Diamond Sectors:** Grants **tax exemptions** to eligible **foreign diamond companies** and **electronics manufacturers**, including income from **rough diamond sales** and **storage of electronic components** in customs bonded warehouses.
- **Tax Exemption for FIIs and BIS:** Exempts **Foreign Institutional Investors (FIIs)** and the **Bank for International Settlements (BIS)** from income tax on **interest and capital gains** from government securities.
- **Easing Rules for Foreign Fund Managers:** The Bill simplifies tax rules to allow **foreign fund managers to relocate to India** without making their foreign funds taxable, while retaining safeguards against **tax avoidance and round-tripping.**
- **Boost for Data Centres:** It removes complex, multi-layered government approval requirements previously mandated for foreign cloud companies to claim tax exemptions when using **Indian data centers.**
  - It also allows Indian data centers to operate on a leased basis rather than strictly under direct ownership, fostering a flexible ecosystem.
- **Tax Benefits for Investment Trusts (REITs & InvITs):** Restores tax-free dividend income for investors in **Real Estate Investment Trusts (REITs) and Infrastructure Investment Trusts (InvITs)** by exempting unit holders, while increasing the surcharge on Special Purpose Vehicles (SPVs) of business trusts from 10% to 25%.
- **UPI Transactions:** Amends the Payment and Settlement Systems Act, 2007 to **empower the Central Government to permit banks and payment service providers to levy charges on Unified**

**Payments Interface (UPI) and other notified electronic payment modes** , replacing the existing prohibition on such charges.

- The measure aims to create a **sustainable revenue model** for banks, **payment service providers (PSPs)** , and payment infrastructure firms while supporting the growth of the digital payments ecosystem.
- The amendments would allow the Central Government to enable the **Merchant Discount Rate (MDR)** on **UPI** and other notified electronic payment modes.
- Currently, banks and payment system providers are prohibited from levying any charges, directly or indirectly, on notified electronic payment modes under **Section 269SU of the Income-tax Act, 1961** .

## Significance

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- Encourages **foreign direct investment (FDI)** and global capital inflows.
- Strengthens the **Make in India initiative** by supporting electronics manufacturing.
- Promotes India as a **global hub for data centres and cloud infrastructure** .
- Positions India as an attractive destination for **global investment fund management** .
- Encourages greater investment in **real estate and infrastructure** through REITs and InvITs.
- Improves **ease of doing business** by reducing regulatory hurdles and providing tax certainty.

# Real Estate Investment Trusts (REITs) & Infrastructure Investment Trusts (InvITs)

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- **REITs:** These are companies that own, operate, or finance income-generating real estate (like office buildings, malls, or hotels).
  - They allow individual investors to earn dividends from real estate investments without having to buy, manage, or finance any properties themselves- similar to how a mutual fund operates.
- **InvITs:** An InvITs is a collective Investment Scheme similar to a mutual fund. InvITs pool money from investors to invest specifically in infrastructure projects (like toll roads, power plants, or transmission lines).
  - InvITs are regulated by the [Securities and Exchange Board of India \(SEBI\) \(Infrastructure Investment Trusts\) Regulations, 2014](#).

## Merchant Discount Rate (MDR)

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- MDR is the fee that a merchant pays to a bank or a payment service provider for accepting payments from customers via digital means (such as debit/credit cards or UPI).
  - It is usually expressed as a percentage of the transaction amount.
- Since **2020** , MDR has remained **zero for UPI and RuPay debit card transactions** to promote digital payments.

## Round-Tripping

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- Round-tripping is a manipulative practice where capital or black money belonging to an Indian entity is illegally transferred to a foreign country (often a tax haven like Mauritius or the Cayman Islands) and is then routed back into India as legitimate Foreign Direct Investment (FDI) or Foreign Portfolio Investment (FPI).

# What are the Concerns Associated Taxation and Other Laws (Amendment) Bill, 2026?

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- **Limited Parliamentary Scrutiny:** The Bill was passed amid disruptions in Parliament with limited debate, raising concerns over legislative scrutiny.
- **Impact on Digital Payments:** The removal of the **Zero- MDR** provision may allow charges on **UPI** transactions, potentially affecting digital payment adoption, especially among **small merchants and low-income users** .
  - However, the move comes amid concerns that the zero-MDR regime has made the UPI ecosystem **financially unsustainable**. With UPI expected to add 600 million users and process 100–150 billion monthly transactions, sustained investment beyond government support will be essential.
- **Revenue Implications:** **Long-term tax exemptions** for foreign companies, **FILs** , and other entities could reduce government tax revenues and affect **fiscal consolidation** .

- **Risk of Round-Tripping:** Relaxed rules for foreign fund managers may increase the risk of **tax avoidance and round-tripping** , despite anti-abuse safeguards.
- **Uneven Competition:** Targeted tax incentives for **foreign companies** may place **domestic Micro, Small and Medium Enterprises (MSMEs)** at a competitive disadvantage.

## Way Forward

- **Link Tax Incentives to Performance:** Make long-term tax exemptions **performance-based** , with **periodic reviews and sunset clauses** linked to outcomes such as **job creation, exports, and technology transfer** .
- **Ensure Data Security:** Align incentives for **foreign cloud companies and data centres** with the **Digital Personal Data Protection (DPDP) Act, 2023** to balance investment promotion with **data protection and cybersecurity** .
- **Strengthen Domestic Manufacturing:** Complement tax incentives with expanded **Production Linked Incentive (PLI) schemes** and **R&D support for MSMEs** to build a resilient domestic electronics supply chain.
- **Prevent Tax Abuse:** Strengthen **SEBI** and **Central Board of Direct Taxes (CBDT)** oversight through **AI-based monitoring** and strict enforcement of the **General Anti-Avoidance Rules (GAAR)** to curb **tax avoidance and round-tripping** .

### ***Drishti Mains Question:***

Relaxation of tax rules for foreign investment funds can attract capital but also raises concerns regarding tax avoidance. Examine.

## Frequently Asked Questions (FAQs)

### **1. What is the objective of the Taxation and Other Laws (Amendment) Bill, 2026?**

It aims to **attract foreign investment, promote Make in India, improve ease of doing business, strengthen digital infrastructure, and rationalise India's tax framework.**

### **2. Which major laws are amended by the Taxation and Other Laws (Amendment) Bill, 2026?**

The Bill amends the **Income-tax Act, 2025** , **Finance Act, 2026** , and **Payment and Settlement Systems Act, 2007** , while replacing the **Income-tax (Amendment) Ordinance, 2026** .

### **3. How does the Bill support electronics manufacturing in India?**

It **extends tax exemptions for foreign companies supplying capital goods to electronics manufacturers** , defines eligible electronic products, and supports the **Make in India** initiative.

### **4. Why are REITs and InvITs important under the Bill?**

The Bill restores **tax-free dividend benefits** for investors in **Real Estate Investment Trusts (REITs)** and **Infrastructure Investment Trusts (InvITs)** , encouraging investment in **real estate and infrastructure** .

### **5. What are the major concerns associated with the Bill?**

Major concerns include **limited parliamentary scrutiny, possible Merchant Discount Rate (MDR) on high-value Unified Payments Interface (UPI) transactions, revenue implications due to tax exemptions, risks of round-tripping, and potential disadvantages for domestic MSMEs.**

# UPSC Civil Services Examination, Previous Year Questions (PYQs)

## Prelims

**Q. Which one of the following effects of creation of black money in India has been the main cause of worry to the Government of India? (2021)**

- (a) Diversion of resources to the purchase of real estate and investment in luxury housing.
- (b) Investment in unproductive activities and purchase of precious stones, jewellery, gold, etc.
- (c) Large donations to political parties and growth of regionalism.
- (d) Loss of revenue to the State Exchequer due to tax evasion.

**Ans: (d)**

## Mains:

**Q.** What is the meaning of the term ‘tax expenditure’? Taking the housing sector as an example, discuss how it influences the budgetary policies of the government. **(2013)**

## GOBARdhan: National Circular Bioenergy Scheme



**Source: PIB**

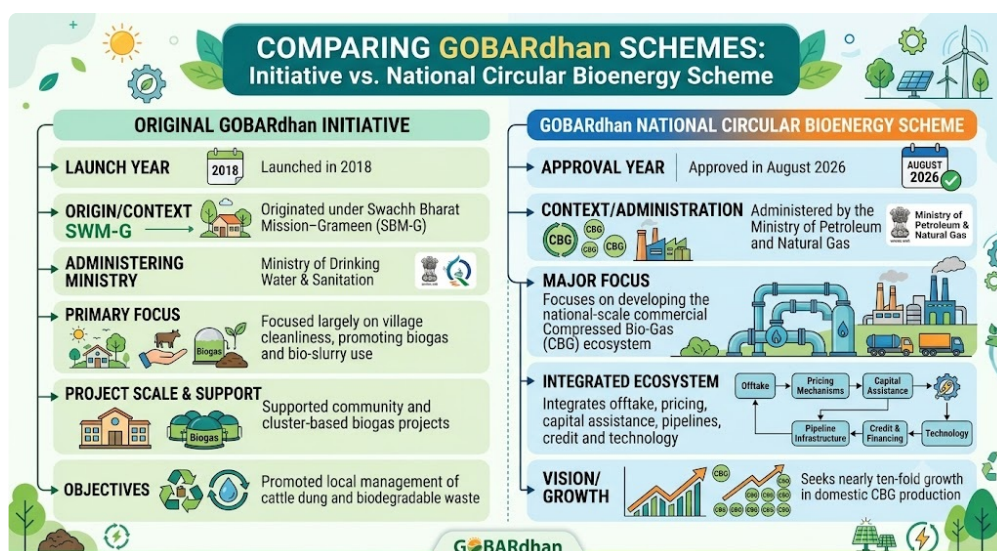
### Why in News?

Recently, the Union Cabinet approved **GOBARdhan , National Circular Bioenergy Scheme** , to convert organic waste into clean energy and promote rural prosperity.

### What are the Key Facts about GOBARdhan National Circular Bioenergy Scheme?

- **About: GOBARdhan, or the National Circular Bioenergy Scheme**, aims to convert organic waste such as **animal waste, crop residue, kitchen waste and market waste** into compressed biogas and bio-slurry.
- **Implementation:** Approved on **6th August 2026** , the scheme will be implemented from **FY 2026-27 to FY 2035-36** under the **Ministry of Petroleum and Natural Gas**.

- **Financial Outlay:** The scheme has a total outlay of ₹23,731 crore and consolidates the previously **fragmented CBG ecosystem** under a unified national framework.
- **Support Framework:** The scheme is **structured around six growth engines** covering assured offtake, pricing, capital support, pipelines, credit guarantees and ecosystem development.
- **Objectives and Vision:** The scheme seeks to **increase domestic CBG production nearly ten-fold, attract private investment and develop a circular bioeconomy** in which waste supports growth, villages become clean-energy centres and farmers contribute to national energy security.
- **Need:** India imports **nearly half of its natural gas requirements, while disruptions around the Strait of Hormuz** —through which about 55-60% of its LNG imports pass —highlight the urgent need to strengthen domestic energy production and security.
- **GOBARDhan Unified Registration Portal:** The **GOBARDhan Unified Registration Portal** enables existing and proposed CBG/Bio-CNG plants to obtain the registration number required for accessing MoPNG support.
  - As of **6th August 2026** , **1,908 plants** were registered, including **217 commissioned plants** producing **0.4 MMSCMD** and **339 plants under construction** .



## Six Growth Engines Under the GOBARDhan Scheme

| Component                                  | Key Provisions                                                                                                                                                                                                                                                                        | Expected Impact                                                                                                                                                            |
|--------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Assured CBG Offtake</b>                 | City Gas Distribution entities will procure CBG in line with the notified obligation of <b>3% in FY 2026-27, 4% in FY 2027-28 and 5% from FY 2028-29 onwards</b> in the CNG-Transport and PNG-Domestic segments.                                                                      | Creates a predictable long-term market, improves project bankability and capacity utilisation, and encourages private investment.                                          |
| <b>Stable CBG Pricing Framework</b>        | Introduces a government-backed administered price of <b>₹2,110 per MMBTU</b> , equivalent to <b>₹105 per kg of CBG</b> , with a minimum <b>10-year horizon</b> and a government-supported, market-based cost-sharing mechanism.                                                       | Provides revenue certainty, remunerative returns and consumer affordability while supporting investment and capacity expansion.                                            |
| <b>Capital Assistance</b>                  | Eligible greenfield projects will receive assistance of up to <b>₹2 crore per tonne per day of installed CBG capacity</b> . Support also covers feedstock aggregation, organic-manure processing and value-addition assets. Capacity-expanding brownfield projects are also eligible. | Reduces initial capital costs, accelerates financial closure and expands participation by private developers, MSMEs, cooperatives and rural entrepreneurs.                 |
| <b>Pipeline Infrastructure Development</b> | Supports cluster-based and standalone pipelines connecting CBG plants with trunk pipelines and City Gas Distribution networks.                                                                                                                                                        | Reduces evacuation costs, improves supply reliability, expands market reach and increases utilisation of domestically produced CBG.                                        |
| <b>Credit Guarantee Support</b>            | Provides credit-guarantee coverage of up to <b>85% on eligible loans</b> for MSME-based CBG projects.                                                                                                                                                                                 | Improves lender confidence, reduces collateral requirements, enables affordable institutional finance and encourages MSMEs, women entrepreneurs and first-time developers. |
| <b>CBG Ecosystem Challenge Fund</b>        | Supports feedstock assessment and mapping, aggregation infrastructure, district-level CBG planning, technology adoption, process improvement, organic-manure value addition, capacity building and stakeholder awareness.                                                             | Accelerates district-level implementation and strengthens local CBG value chains and ecosystem capabilities.                                                               |

## What Are the Expected Outcomes of GOBARdhan

# Scheme?

- **Private Investment:** Assured offtake, stable pricing, capital assistance and improved institutional finance will **enhance project viability** and attract large-scale private investment.
- **Economic Contribution:** Expansion of the CBG industry is expected to contribute more than **₹75,000 crore to India's GDP** over ten years.
- **Employment and Rural Livelihoods:** The sector is projected to generate over **1.5 lakh jobs** and create additional income opportunities for farmers, feedstock aggregators, cooperatives and rural entrepreneurs.
- **Scientific Waste Management:** Agricultural residue, **cattle dung, municipal organic waste and other biomass resources** will be productively converted into clean fuel and organic manure, reducing unmanaged waste and landfill disposal.
- **Climate Benefits:** Replacement of fossil fuels and diversion of organic waste from landfills are expected to reduce more than **40 MT of CO<sub>2</sub> emissions** .
- **Organic-Manure Economy:** The Scheme is expected to produce around **250 MMT of organic fertiliser** , while expanding the production, value addition and use of **Fermented Organic Manure and Liquid Fermented Organic Manure** .

## Government Initiatives for Building CBG Ecosystem

- **Foundational Policy Support:** India has developed the CBG ecosystem through the **SATAT initiative** , **Market Development Assistance Scheme** , **Biomass Aggregation Machinery Scheme** , **Development of Pipeline Infrastructure Scheme** and **Central Financial Assistance under the National Bioenergy Programme** .
- **Biomass Aggregation Support:** The **BAM Scheme** , implemented by the **Ministry of Petroleum and Natural Gas** , helps CBG producers procure biomass-collection equipment. It has an outlay of **₹564.75 crore** , with 37 proposals involving nearly **₹248 crore** approved by 16 March 2026.
- **Pipeline Connectivity:** The **DPI Scheme** provides **₹994.5 crore** during FY 2024-25 to FY 2028-29 to strengthen CBG evacuation and market access. So far, **₹56.31 crore** has been sanctioned.
- **Organic-Manure Promotion:** The **Market Development Assistance (MDA) Scheme** provides **₹1,500 per metric tonne** for Fermented Organic Manure, Liquid Fermented Organic Manure and Phosphate-Rich Organic Manure produced by GOBARdhan and CBG plants.
  - As of 4th March 2026, 120 CBG/biogas plants were registered on the fertiliser portal and 44 MoUs had been signed. Between February 2025 and January 2026, 15,690 Kisan Sangoshthis and awareness camps were organised.

## Conclusion

GOBARdhan seeks to transform **India's organic waste into clean energy, rural income and economic value** by integrating assured demand, stable pricing, capital support, pipeline connectivity and credit assistance. By scaling up the CBG ecosystem, it advances **energy security,**

## Frequently Asked Questions

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### 1. What is the full form of GOBARdhan?

GOBARdhan stands for **Galvanizing Organic Bio-Agro Resources Dhan** .

### 2. When was the original GOBARdhan initiative launched?

It was launched on 30th April 2018 under the Swachh Bharat Mission–Grameen framework.

### 3. What is the CBG Obligation?

It requires CBG procurement or blending in the CNG transport and domestic PNG segments at 3% in 2026–27, 4% in 2027–28 and 5% from 2028–29 onwards.

### 4. Which ministry administers the new scheme?

The scheme is administered by the **Ministry of Petroleum and Natural Gas** .

### 5. What is compressed biogas?

CBG is biogas purified to near-natural-gas quality and compressed for use as a fuel. It can be integrated into the existing natural-gas ecosystem.

[Watch Video on YouTube:

▶ <https://www.youtube.com/embed/zCcTPXCkNvA>

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# UPSC Civil Services Examination, Previous Year Questions

## Prelims

**Q. According to India's National Policy on Biofuels, which of the following can be used as raw materials for the production of biofuels? [2020]**

1. Cassava
2. Damaged wheat grains
3. Groundnut seeds
4. Horse gram
5. Rotten potatoes
6. Sugar beet

**Select the correct answer using the code given below:**

- a) 1, 2, 5 and 6 only
- b) 1, 3, 4 and 6 only
- c) 2, 3, 4 and 5 only
- d) 1, 2, 3, 4, 5 and 6

**Ans: (a)**

## Mains

**Q.** What are the impediments in disposing of the huge quantities of discarded solid wastes which are continuously being generated? How do we safely remove the toxic wastes that have been accumulating in our habitable environment? **[2018]**

# Veeragallu Discovery Sheds Light on Ruler Vishnuvardhana's Death



**Source:TH**

A **12th-century Veeragallu (hero stone)** bearing an **Old Kannada inscription** unearthed in **Neeragunda village, Karnataka** , may help resolve the long-standing historical debate over the **death of Hoysala ruler Vishnuvardhana** , making it an important archaeological discovery.

- **Discovery Site:** The **Veeragallu** was discovered during restoration work at the **Hoysala-era Lakshmi Narayana Swamy Temple in Neeragunda, Karnataka** .

- **Inscription Details:** The inscription states that Maravve erected the Veeragallu in **memory of her husband, Boppa, a Hoysala soldier** , who died shortly after the **death of King Vishnuvardhana at Bankapura** (present-day Haveri district).
  - It records the date as Ekadashi of Pushya Masa, Durmathi Samvatsara, Year 1064, corresponding to **11th December 1141 CE (Gregorian Calendar)**.
  - If authenticated, this inscription could resolve the long-standing dispute regarding Vishnuvardhana's death, previously believed by different scholars to have occurred in 1149 CE or 1152 CE.
- **About Veeragallu (Hero Stones): Memorial stones** erected to honour individuals who died defending their **king, village, or cattle** .
  - Usually feature **three panels** depicting the **battle scene, ascent to heaven, and worship before a deity** , symbolising **Moksha (salvation)** .
  - Reflect the Hoysala **Garuda tradition** , where elite royal bodyguards displayed absolute loyalty and, in some cases, sacrificed their lives following the king's death.
- **King Vishnuvardhana:** Originally known as **Bitti Deva** , he was one of the **greatest rulers of the Hoysala dynasty** .
  - **Defeated the Cholas in the Battle of Talakad (1116 CE)** , consolidating Hoysala power in southern India.
  - The Chennakeshava Temple at Belur, Karnataka, is one of the finest masterpieces of Hoysala architecture. It was consecrated by the Hoysala ruler **Vishnuvardhana** in **1117 CE** to commemorate his **1116 CE victory over the Cholas** and was originally known as the **Vijaya Narayana Temple** .
  - Influenced by **Ramanujacharya** , he embraced **Srivaishnavism** while continuing to patronise **Jainism and Shaivism** .
- **About the Hoysala Empire:** Ruled parts of present-day **Karnataka, Tamil Nadu, and Andhra Pradesh between the 11th and 14th centuries CE**.
  - Capitals were **Belur** and later **Halebidu (Dwarasamudra)** .
  - **Hoysala temples** are renowned for their **distinctive Vesara-style architecture** , characterized by **complex stellate (star-shaped) plans, raised circumambulatory platforms (Jagati), tiered friezes, and intricate soapstone carvings** .
  - The **Sacred Ensembles of the Hoysalas (Belur, Halebidu, and Somanathapura)** were inscribed as a UNESCO World Heritage Site in 2023.

**Read more: Hoysala Temples**

- **Menhirs**

# India Successfully Test-fires Agni-4 Ballistic Missile



## Source: TH

Recently, India successfully conducted the test launch of the **Agni-4 medium-range ballistic missile** from the **Integrated Test Range (ITR), Chandipur, Odisha** .

- The launch was carried out under the aegis of the **Strategic Forces Command** , which manages **India's strategic nuclear forces** .
- **About:** Agni-4 is an indigenously developed, **intermediate-range, surface-to-surface ballistic missile** belonging to India's Agni missile series.
- **Features:** Developed by the **Defence Research and Development Organisation (DRDO)** , Agni-IV is approximately 20 metres long, has a strike range of about 4,000 km and can carry a payload of around 1,000 kg.
  - The missile uses **advanced avionics, onboard computers, inertial navigation and guidance systems** to achieve high accuracy and maintain stability during flight.
- **Significance:** The successful test demonstrated Agni-4's operational readiness and reinforced **India's credible strategic-deterrence capability** .

# AGNI MISSILE SERIES BREAKDOWN

## AGNI-IV

Advanced IRBM

RANGE  
**3,500–  
4,000 km**

PURPOSE | Tactical nuclear deterrence  
against neighbouring threats

TECHNOLOGY | Ring laser gyro and composite  
rocket motor

CAPABILITY | Nuclear-capable precision strike missile



## AGNI-V

INTERCONTINENTAL BALLISTIC  
MISSILE (ICBM)

RANGE  
**5,000+ km**

STAGES | Three-stage solid-fuel  
missile

TECHNOLOGY | Canister launch system  
and MIRV capability

KEY FEATURE | Faster launch readiness and  
multiple warhead delivery

STATUS | Successfully tested with MIRV technology in  
2024 and 2025



## AGNI-P (Prime)

NEXT-GENERATION  
BALLISTIC MISSILE

RANGE  
**1,000–  
2,000 km**  
(estimated)

PURPOSE | Replacement for  
Agni-I and Agni-II

TECHNOLOGY | Lighter composite  
body with advanced guidance

KEY FEATURE | High accuracy and dual-  
canister launch capability



# Tanvi Sharma Wins Taipei Open Super 300



Source: TH

Recently, **Indian shuttler Tanvi Sharma won** her maiden **BWF World Tour title at the Taipei Open Super 300**.

- **About:** Aged 17 years and 222 days, she became the **youngest finalist in the tournament's history**, surpassing Lee Yong Dae's 2006 record.
- **Historic Achievement:** Coached by Park Tae-sang, former coach of P.V. Sindhu, Tanvi became the **first Indian woman** to win the Taipei Open singles title since Saina Nehwal in 2008.
- **Elite Indian Group:** The triumph placed her among the few Indian women—alongside Saina Nehwal, P.V. Sindhu and Devika Sihag, to win a BWF World Tour Super 300 title.
- **Junior Achievements:** A former World Junior No. 1, Tanvi became the **first Indian to win two medals at a single edition** of the World Junior Championships—silver in girls' singles and bronze in the mixed-team event.
- **Other Achievements:** She finished runner-up at the 2025 US Open Super 300 and was part of India's gold medal-winning team at the Asian Team Championships.
- **BWF World Tour Super 300:** It is a Level 5 international badminton tournament under the BWF's Grade 2 competition structure, **ranked below Super 1000, Super 750 and Super 500 events**. It offers players world-ranking points and prize money.



**Read More:** [PV Sindhu Becomes First Indian to Win Japan Open Title](#)